

On the Partition Algebra of Bounded Phase Space: Composition-Inflation, Angular Resolution, and the Categorical Reformulation of Fundamental Constants

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Abstract

Every persistent physical system occupies a bounded region of phase space. From this single empirical observation, promoted to an axiom, we derive by deduction the complete partition algebra governing distinguishable states in bounded oscillatory systems.

The central result is the *composition-inflation formula*: a bounded oscillatory system with d entropy dimensions completing n oscillation cycles generates

$$T(n, d) = d \cdot (d + 1)^{n-1}$$

categorically distinguishable trajectories — an exponential, not linear, count. The formula is exact, has no adjustable parameters, and follows from the binomial theorem applied to labeled integer compositions.

We derive three consequences that are not available in any existing framework. First, the *angular resolution* $\Delta\theta = 2\pi/T(n, d)$ is dimensionless and therefore not subject to the Planck time constraint, which bounds temporal intervals measured in seconds but does not bound dimensionless ratios. For any $\varepsilon > 0$ there exists a finite n achieving $\Delta\theta < \varepsilon$. Second, all fundamental constants except the gravitational constant G reduce to tick counts, factors of 2π , and a single energy unit E_{tick} in categorical angular units: $c = 2\pi \text{ rad/tick}$, $\hbar = E_{\text{tick}}/(2\pi)$, $k_B = E_{\text{tick}}/\ln(d + 1)$. Third, G is the unique irreducible coupling constant — the only physical constant not expressible in terms of counting and geometry.

The paper is self-contained: all results are proved from the single axiom with no additional physical postulates. Applications to the LHC trigger, mass spectrometry, and gravitational physics are indicated in the final section but not developed here.

Keywords: bounded phase space; partition algebra; composition-inflation; angular resolution; Planck depth; categorical angular units; gravitational constant

Contents

1 Introduction

1.1 The single empirical premise

Physical systems exist. Every physical system that persists — an electron bound to a nucleus, a molecule in a container, a planet in a stellar orbit, an ion in a trap — occupies a bounded region of phase space. The complement — a system whose phase space region grows without bound — disperses and ceases to persist. This is not a theoretical proposal; it is a summary of every observation of every persistent object in the history of physics.

We promote this observation to an axiom and derive consequences.

1.2 What follows

The derivation proceeds in four steps. Section ?? states the axiom and derives oscillatory necessity (Poincaré recurrence eliminates every alternative). Section ?? constructs the partition coordinate system (n, ℓ, m, s) with capacity $C(n) = 2n^2$ from the geometry of bounded oscillation. Section ?? derives the composition-inflation formula $T(n, d) = d(d + 1)^{n-1}$ and establishes the angular resolution theorem. Section ?? expresses all fundamental constants in categorical angular units, identifies G as the unique irreducible constant, and derives the Planck depth.

No theorem in Sections ??–?? uses any postulate beyond the axiom of Section ??.

2 Bounded Phase Space and Oscillatory Necessity

Axiom 1 (Bounded Phase Space). Every persistent dynamical system (Ω, μ, ϕ_t) satisfies:

- (i) $\Omega \subset \mathbb{R}^{2d}$ is bounded with finite Liouville measure $\mu(\Omega) < \infty$.
- (ii) $\phi_t : \Omega \rightarrow \Omega$ preserves μ for all t .
- (iii) Ω admits a sequence of finite measurable partitions $\mathcal{P}_1 \preceq \mathcal{P}_2 \preceq \dots$ with each $\mathcal{P}_{k+1}$ refining $\mathcal{P}_k$ and

the generated σ -algebra equalling the Borel algebra modulo null sets.

Theorem 2.1 (Poincaré Recurrence ?). *Under Axiom ??, for every measurable $A \subset \Omega$ with $\mu(A) > 0$ and every $T > 0$, almost every $x \in A$ satisfies $\phi_{t_n}(x) \in A$ for infinitely many $t_n \rightarrow \infty$.*

Proof. Suppose not. Let $B = \{x \in A : \phi_t(x) \notin A \forall t > T\}$. The iterates $\{\phi_{-kT}(B)\}_{k \geq 0}$ are pairwise disjoint (if $x \in \phi_{-jT}(B) \cap \phi_{-kT}(B)$ with $j < k$, then $\phi_{(k-j)T}(x) \in B$ and later $\phi_t(x)$ returns to A , contradicting $x \in B$). Measure preservation gives $\sum_{k=0}^{\infty} \mu(B) = \mu(\bigcup_k \phi_{-kT}(B)) \leq \mu(\Omega) < \infty$, so $\mu(B) = 0$. $\square$

Corollary 2.2 (Oscillatory Necessity). *Under Axiom ??, every persistent non-static trajectory is oscillatory: no coordinate can grow monotonically, and every trajectory returns to every neighbourhood of its starting point infinitely often.*

Proof. Monotonic growth contradicts boundedness. Absence of recurrence contradicts Theorem ??. The only consistent dynamics is recurrent oscillatory motion. $\square$

For integrable systems, the Arnold–Liouville theorem (?) guarantees action-angle variables (J_i, θ_i) with quasi-periodic motion $\dot{\theta}_i = \omega_i(J)$. The Koopman operator (?) on $L^2(\Omega, \mu)$ has a pure point spectrum on cyclic subspaces; its eigenfrequencies $\{\omega_k\}$ are the mode frequencies of the system. Near-integrable systems retain this structure on most tori by the KAM theorem (???).

3 Partition Coordinates and Shell Capacity

3.1 Hierarchical partition and coordinate generation

The hierarchical partition of Axiom ??(iii) is not merely formal. For a three-dimensional bounded system with $SO(3)$ symmetry, the refinement sequence generates a natural four-coordinate label on each partition cell.

Definition 3.1 (Partition State). A *partition state* of a bounded three-dimensional Hamiltonian system is an equivalence class of phase-space trajectories sharing the same closure label (n, ℓ, m, s) where: $n \in \{1, 2, 3, \dots\}$ (principal level, radial closure count), $\ell \in \{0, 1, \dots, n-1\}$ (angular sublevel, polar closure count), $m \in \{-\ell, \dots, +\ell\}$ (azimuthal projection), $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ (rotational parity). The set of all partition states is $\mathcal{P}$.

The label (n, ℓ, m, s) is derived from the closure conditions of orbits in action-angle coordinates. An orbit closes when $n_r \omega_r = n_\theta \omega_\theta = n_\phi \omega_\phi$ for positive integers; the hierarchy of radial, polar, and azimuthal boundaries produces the nesting $0 \leq \ell \leq n-1$, $|m| \leq \ell$, and the binary parity s from the double cover $\text{Spin}(3) \cong SU(2) \rightarrow SO(3)$ (?).

Theorem 3.2 (Shell Capacity). *The number of distinct partition states at principal level n is*

$$C(n) = 2n^2. \quad (1)$$

Proof. At level n , ℓ ranges over $\{0, \dots, n-1\}$; for each ℓ , m ranges over $2\ell+1$ values; s contributes a factor of 2:

$$C(n) = 2 \sum_{\ell=0}^{n-1} (2\ell+1) = 2(n + 2 \cdot \frac{n(n-1)}{2}) = 2n^2. \quad \square \quad (2)$$

Corollary 3.3 (Cumulative Count). *The total number of partition states at or below level $n_{\max}$ is*

$$N_{\text{state}}(n_{\max}) = \sum_{n=1}^{n_{\max}} 2n^2 = \frac{n_{\max}(n_{\max}+1)(2n_{\max}+1)}{3}. \quad (3)$$

Remark 3.4. The sequence $C(n) = 2, 8, 18, 32, 50, 72, 98, \dots$ is the electron shell capacity sequence of the periodic table. This is not coincidental: both arise from the same $SO(3)$ representation theory applied to a bounded three-dimensional oscillatory system.

3.2 The partition bijection

Theorem 3.5 (Partition Bijection). *There exists a bijection $\Phi: \mathbb{Z}^+ \rightarrow \mathcal{P}$ mapping each positive integer M to a unique partition state (n, ℓ, m, s) . The map is constructive: given M , the principal level is $n = \lfloor \sqrt{M/2} \rfloor + 1$ (adjusted to the correct shell), ℓ and m follow from the lexicographic ordering within the shell, and $s = \pm \frac{1}{2}$ from the parity index.*

Proof. Injectivity: distinct M values yield distinct shell positions in the lexicographic ordering, hence distinct (n, ℓ, m, s) tuples. Surjectivity: every valid tuple (n, ℓ, m, s) has a unique cumulative index $M = N_{\text{state}}(n-1) + \delta$ where δ is the within-shell offset; this M maps back to (n, ℓ, m, s) by construction. $\square$

Theorem 3.6 (Strict Monotonicity of the Partition Count). *Let $M(t)$ be the partition count of a bounded oscillatory system with oscillation frequency $f > 0$. Then $M(t) = ft$ exactly, and $M(t_2) > M(t_1)$ whenever $t_2 > t_1$ and at least one cycle has completed.*

Proof. Each completed oscillation cycle increments M by exactly 1. The rate is $dM/dt = f = \omega/(2\pi)$, exact by definition of the oscillation count. Strict monotonicity follows since $f > 0$. $\square$

4 Composition-Inflation and Angular Resolution

4.1 Integer compositions

Definition 4.1 (Composition). An *integer composition* of n is an ordered sequence $(c_1, \dots, c_k)$ of positive inte-

gers summing to n . Two compositions differing in order are distinct.

Theorem 4.2 (Composition Count ?). *The total number of compositions of n is 2^{n-1} .*

Proof. Place n objects in a row; there are $n - 1$ gaps. Each gap either contains a divider or does not. The 2^{n-1} binary choices on the gaps are in bijection with compositions. $\square$ $\square$

The number of compositions into exactly k parts is $\binom{n-1}{k-1}$.

4.2 Dimensional labeling

The S-entropy space is the unit cube $[0, 1]^d$ with d entropy dimensions. For $d = 3$ (knowledge S_k , temporal S_t , evolution S_e), any oscillatory trajectory through this space can be characterized by which dimension it refines at each step.

Definition 4.3 (Labeled Composition). A *labeled composition* of n in d dimensions is a pair $(\mathbf{C}, \mathbf{L})$ where $\mathbf{C} = (c_1, \dots, c_k)$ is a composition of n and $\mathbf{L} = (\ell_1, \dots, \ell_k)$ with $\ell_i \in \{1, \dots, d\}$ specifies the entropy dimension refined during the i -th part of the trajectory.

Theorem 4.4 (Composition-Inflation Formula). *The total number of labeled compositions of n in d dimensions is*

$$T(n, d) = d \cdot (d + 1)^{n-1}. \quad (4)$$

Proof. Summing over all numbers of parts:

$$T(n, d) = \sum_{k=1}^n \binom{n-1}{k-1} d^k = d \sum_{j=0}^{n-1} \binom{n-1}{j} d^j = d(1+d)^{n-1} \quad (5)$$

where the last step is the binomial theorem with $x = d$, $m = n - 1$. $\square$

Proposition 4.5 (Growth Properties). $T(1, d) = d$; $T(n + 1, d) = (d + 1) \cdot T(n, d)$ (each additional cycle multiplies by $d + 1$); $\log T(n, d) = \log d + (n - 1) \log(d + 1)$; $T(n, d)/n \rightarrow \infty$ as $n \rightarrow \infty$.

Corollary 4.6 (S-Entropy Space). For $d = 3$: $T(n, 3) = 3 \cdot 4^{n-1}$. After $n = 56$ cycles, $T(56, 3) = 3 \cdot 4^{55} \approx 3.8 \times 10^{33}$.

4.3 Trajectory distinguishability

Theorem 4.7 (Position-Trajectory Duality). *In the ternary encoding of the unit cube $[0, 1]^3$, the trit string encodes both the final position and the trajectory taken to reach it. Two distinct labeled compositions produce distinct trit strings and are therefore categorically distinguishable.*

Proof. Case 1: distinct underlying compositions. Different sequences of segment lengths determine transitions at different depths, yielding distinct trit strings. Case

2: same composition, distinct labels. There exists index i with $\ell_i \neq \ell'_i$; the i -th segment refines different axes, producing distinct trit strings by orthogonality of the dimensions. Both cases yield distinct categorical states. $\square$ $\square$

Theorem 4.8 (Non-Demolition Distinguishability). *Let $\hat{O}_{\text{cat}}$ measure categorical trajectory labels and $\hat{O}_{\text{phys}}$ measure any physical observable (position, momentum, energy). Then $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$. Measuring which trajectory was taken does not disturb the physical state.*

Proof. $\hat{O}_{\text{cat}}$ acts on the trajectory label — a topological property of the path. $\hat{O}_{\text{phys}}$ acts on the instantaneous phase-space position. These act on complementary degrees of freedom: the trajectory label is constant along the trajectory (a conserved quantity), hence commutes with the Hamiltonian and with all physical observables it generates. Since $\hat{O}_{\text{cat}}$ is diagonal in the trajectory basis and $\hat{O}_{\text{phys}}$ acts on the physical space factor, their tensor product structure guarantees $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$. $\square$ $\square$

4.4 Angular resolution and the Planck constraint

Definition 4.9 (Angular Resolution). For $T(n, d)$ distinguishable trajectories distributed uniformly over $[0, 2\pi)$, the angular resolution is

$$\Delta\theta(n, d) = \frac{2\pi}{T(n, d)} = \frac{2\pi}{d(d+1)^{n-1}}.$$

Theorem 4.10 (No Planck Limit on Angular Resolution). *The Planck time $t_P = \sqrt{\hbar G/c^5} \approx 5.39 \times 10^{-44}$ s bounds temporal intervals measured in seconds. The angular resolution $\Delta\theta(n, d) = 2\pi/T(n, d)$ is dimensionless (units of radians) and is therefore not bounded below by any function of t_P . For any $\varepsilon > 0$, there exists*

$$n_0 = 1 + \left\lceil \log_{d+1} \left(\frac{2\pi}{d\varepsilon} \right) \right\rceil$$

such that $\Delta\theta(n, d) < \varepsilon$ for all $n \geq n_0$.

Proof. t_P has dimension $[t_P] = \text{s}$. The Heisenberg bound $\Delta E \cdot \Delta t \geq \hbar/2$ constrains energy–time products; both factors carry physical dimensions. $\Delta\theta = 2\pi/T(n, d)$ is the ratio of two dimensionless quantities and carries no time dimension: $[\Delta\theta] = \text{rad} = 1$. Neither the Heisenberg argument nor the gravitational collapse argument applies to a dimensionless ratio.

Since $T(n, d) = d(d + 1)^{n-1}$ is unbounded, $\Delta\theta \rightarrow 0$ as $n \rightarrow \infty$. Setting $\Delta\theta < \varepsilon$ and solving gives n_0 as stated. $\square$

Remark 4.11. The composition-inflation mechanism does not create shorter time intervals. It creates more distinguishable trajectory structures within the same time interval. The Planck limit bounds temporal duration; composition-inflation operates on categorical multiplicity. These are fundamentally different quantities.

4.5 The Planck depth

Definition 4.12 (Planck Depth). The *Planck depth* n_P for a reference oscillator with period τ_{osc} in d -dimensional entropy space is the minimum cycle count for which the categorical trajectory count equals or exceeds the number of Planck-time intervals in one oscillator period:

$$T(n_P, d) \geq \frac{\tau_{\text{osc}}}{t_P}.$$

Theorem 4.13 (Planck Depth Formula).

$$n_P = 1 + \left\lceil \log_{d+1} \left(\frac{\tau_{\text{osc}}}{d \cdot t_P} \right) \right\rceil.$$

Proof. Solve $d(d+1)^{n_P-1} \geq \tau_{\text{osc}}/t_P$ for n_P . $\square$ $\square$

Theorem 4.14 (Universality of Planck Depth). *For any physically realizable oscillator with frequency ν satisfying $t_P^{-1} > \nu > 1$ Hz, the Planck depth at $d = 3$ lies in $n_P \in [36, 73]$. The variation spans only 37 cycles despite frequencies varying over 14 orders of magnitude, because n_P depends logarithmically on ν .*

Proof. At the lower frequency extreme ($\nu = 1$ Hz): $\tau/t_P = 1.86 \times 10^{43}$, $n_P = 1 + \lceil \log_4(1.86 \times 10^{43}/3) \rceil = 73$. At the strontium optical clock ($\nu = 4.29 \times 10^{14}$ Hz): $\tau/t_P = 4.32 \times 10^{28}$, $n_P = 48$. The logarithm compresses 14 orders of magnitude in frequency to $73 - 48 = 25$ cycles. $\square$

Table 1: Planck depth for representative oscillators ($d = 3$).

Oscillator	ν (Hz)	τ/t_P	n_P
Sr optical clock	4.29×10^{14}	4.3×10^{28}	48
Cs-133 hyperfine	9.19×10^9	2.0×10^{33}	56
LHC bunch clock	4.01×10^7	4.6×10^{35}	60
H maser	1.42×10^9	1.3×10^{34}	57
CPU (3 GHz)	3.00×10^9	6.2×10^{33}	57

5 Categorical Angular Units and Constant Reduction

5.1 The speed of light

Theorem 5.1 (Speed of Light as Angular Velocity). *In categorical angular units, the speed of light is exactly*

$$c = 2\pi \text{ rad/tick}. \quad (6)$$

Proof. In one oscillation cycle (one tick), a bounded system traverses the full phase $\Delta\phi = 2\pi$ rad of its reference oscillation. The maximum rate at which categorical distinctions propagate between two points separated by Δx at the characteristic radius r_0 is $c = \Delta x/\tau_c = 2\pi r_0/\tau_{\text{tick}}$ where $\tau_{\text{tick}} = 1$ tick. In angular units $\theta = x/r_0$: $c = 2\pi \text{ rad/tick}$.

The partition propagation bound establishes that $c = \sup_n \ell_n \omega_n$ is finite and equals this value; the bound is attained at the massless (zero-rest-frequency) limit. $\square$

5.2 Planck's constant

Theorem 5.2 (Planck's Constant as Energy per Radian).

$$\hbar = E_{\text{tick}}/(2\pi), \quad (7)$$

where E_{tick} is the energy carried by one oscillation tick.

Proof. From $E = \hbar\omega$ with $\omega = 2\pi/\tau_{\text{tick}} = 2\pi \text{ tick}^{-1}$: $E_{\text{tick}} = \hbar \cdot 2\pi$. Solving: $\hbar = E_{\text{tick}}/(2\pi)$. $\hbar$ is the energy per radian of phase advance. $\square$

5.3 Boltzmann's constant

Theorem 5.3 (Boltzmann's Constant as Entropy per Composition Step).

$$k_B = E_{\text{tick}}/\ln(d+1). \quad (8)$$

Proof. By Proposition ??(ii), each additional oscillation tick multiplies the trajectory count by $(d+1)$, contributing $\ln(d+1)$ natural units of categorical entropy. The thermal energy per degree of freedom is $k_B T$, where T is the tick rate. The energy per unit of composition entropy is $E_{\text{tick}}/\ln(d+1)$. For $d = 3$: $k_B = E_{\text{tick}}/\ln 4$. $\square$

5.4 Mass and rest energy

Corollary 5.4 (Mass and Rest Energy). *The inertial mass and rest energy of a localized oscillator with rest frequency ν_0 (ticks^{-1}) are*

$$m_0 = \frac{E_{\text{tick}} \cdot \nu_0}{4\pi^2}, \quad (9)$$

$$E_0 = m_0 c^2 = E_{\text{tick}} \cdot \nu_0. \quad (10)$$

Proof. From $m_0 = \hbar\omega_0/c^2$ with $\hbar = E_{\text{tick}}/(2\pi)$ and $c = 2\pi \text{ rad/tick}$: $m_0 = E_{\text{tick}}\omega_0/(2\pi \cdot 4\pi^2) = E_{\text{tick}}\nu_0/(4\pi^2)$ where $\nu_0 = \omega_0/(2\pi)$. Then $E_0 = m_0 c^2 = E_{\text{tick}}\nu_0/(4\pi^2) \cdot 4\pi^2 = E_{\text{tick}}\nu_0$. The identity $E_0 = \hbar\omega_0$ is recovered: rest energy is the fundamental frequency-energy identity applied to the localization mode. $\square$

5.5 The Planck time and the irreducibility of G

Theorem 5.5 (Planck Time Decomposition).

$$t_P = \sqrt{\hbar G/c^5} = \frac{\sqrt{E_{\text{tick}} \cdot G}}{(2\pi)^3}. \quad (11)$$

The Planck time has two components: geometric factors $(2\pi)^3$ from the angular structure of oscillation, and the product $\sqrt{E_{\text{tick}} \cdot G}$ as the unique irreducible coupling.

Proof. Substitute $\hbar = E_{\text{tick}}/(2\pi)$ and $c = 2\pi \text{ rad/tick}$: $t_P = \sqrt{E_{\text{tick}}G/(2\pi \cdot (2\pi)^5)} = \sqrt{E_{\text{tick}}G}/(2\pi)^3$. $\square$

Theorem 5.6 (Irreducibility of G). *The gravitational constant G is the unique fundamental constant of physics that does not reduce to a combination of tick counts, factors of 2π , and the energy unit E_{tick} .*

Proof. Table ?? shows the reduction of six of the seven fundamental constants. The remaining constant G appears only in t_P and the Schwarzschild radius; in both cases it appears as an irreducible multiplicative factor that cannot be expressed in terms of counting operations or geometry. It encodes the coupling between categorical partition structure (which determines mass through $m_0 = E_{\text{tick}}\nu_0/(4\pi^2)$) and the curvature of space. No counting argument reduces this coupling. $\square$

Table 2: Fundamental constants in categorical angular units.

Constant	SI value	Categorical form
c	$3.00 \times 10^8 \text{ m/s}$	$2\pi \text{ rad/tick}$
$\hbar$	$1.055 \times 10^{-34} \text{ J} \cdot \text{s}$	$E_{\text{tick}}/(2\pi)$
k_B	$1.381 \times 10^{-23} \text{ J/K}$	$E_{\text{tick}}/\ln(d+1)$
m_0	$\hbar\omega_0/c^2$	$E_{\text{tick}}\nu_0/(4\pi^2)$
E_0	m_0c^2	$E_{\text{tick}}\nu_0 = \hbar\omega_0$
t_P	$5.39 \times 10^{-44} \text{ s}$	$\sqrt{E_{\text{tick}}G}/(2\pi)^3$
G	$6.674 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}$	irreducible

Corollary 5.7 (Planck Depth from G). *Since $t_P = \sqrt{E_{\text{tick}}G}/(2\pi)^3$ and n_P depends on t_P through Theorem ??, the Planck depth is computable to precision $(d+1)^{-n}$ once E_{tick} and G are known. Since G is the unique irreducible constant, the Planck depth is the one quantity in the entire categorical framework that is not derivable from geometry and counting alone. It is the fingerprint of gravity in the composition-inflation structure.*

6 Discussion and Outlook

6.1 What has been established

The paper has derived, from a single empirical axiom:

- (1) Oscillatory necessity — bounded dynamics must oscillate (Corollary ??).
- (2) The partition coordinate system (n, ℓ, m, s) with capacity $C(n) = 2n^2$ (Theorem ??).
- (3) The partition bijection $\Phi : \mathbb{Z}^+ \rightarrow \mathcal{P}$ and strict monotonicity of the partition count (Theorems ??, ??).
- (4) The composition-inflation formula $T(n, d) = d(d+1)^{n-1}$ (Theorem ??).
- (5) Angular resolution is dimensionless and Planck-unconstrained (Theorem ??).
- (6) The Planck depth formula and its universality across oscillators (Theorems ??, ??).
- (7) All fundamental constants except G reduce to 2π , tick counts, and E_{tick} (Theorems ??-?? and Table ??).
- (8) G is the unique irreducible constant (Theorem ??).

6.2 Applications developed in companion papers

The partition algebra derived here is the mathematical foundation for a sequence of companion papers. The following applications are developed there, citing only results proved above:

LHC trigger design. A trigger system with d independent detector subsystems observing n bunch crossings can discriminate $T(n, d)$ distinct event categories. The Planck depth $n_P = 60$ for the LHC bunch clock ($\nu_B = 40 \text{ MHz}$, $d = 3$) defines a natural design depth; the current L1 trigger at $n = 100$ operates at $n/n_P = 1.67$, placing it in the super-Planck categorical resolution regime (??).

Mass spectrometry. The partition Lagrangian $\mathcal{L}_{\mathcal{M}} = \frac{1}{2}\mu|\dot{\mathbf{x}}|^2 + \mu\dot{\mathbf{x}} \cdot \mathbf{A}_{\mathcal{M}} - \mathcal{M}(\mathbf{x}, t)$ derives all four mass analyzer equations (TOF, quadrupole, Orbitrap, FT-ICR) from the same Euler-Lagrange equations, with the Lorentz force as a consequence rather than a postulate (?).

Gravitational constant. Three independent routes to G via the partition algebra — oscillation-ratio, category fixed-point, and partition-density ratio — converge on the same value with $(d+1)^{-n}$ precision. At $n = 8$ all three agree with CODATA; at $n = 27$ they reach double-precision machine epsilon (?).

Cosmology. A partition-density-dependent G reproduces the MOND acceleration scale $a_0 = cH_0/(2\pi)$, predicts dark energy equation of state $w_{\text{eff}} = -1 + 1/(d+1) = -0.75$, and secular drift $\dot{G}/G = -3H_0/(d+1)$ (?).

6.3 Falsification conditions

The framework is falsified by any of the following:

- A persistent physical system with infinite Liouville measure.
- A bounded oscillatory system in which the partition count $M(t)$ decreases (i.e., a partition decrement is physically realized).
- A physical observable that cannot be expressed as a function of $(n, \ell, m, s, E_{\text{tick}}, G)$.

None of these conditions has been observed.

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